Sun et al. BMC Geriatrics (2025) 25:639
https://doi.org/10.1186/s12877-025-06304-x

BMC Geriatrics

RESEARCH

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Factors that influence technophobia in Chinese older patients with ischemic stroke: a cross-sectional survey

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Abstract

Background Older patients with ischemic stroke often have a large number of medical needs, technophobia refers to the irrational anxiety and fear of digital technologies such as mobile communication equipment, artificial intelligence and robots, resulting in emotional reactions such as fear and tension, which will lead individuals to avoid using technology. However, the research on these patients' technophobia and the factors that may lead to it is limited.

Aims This study aimed to investigate the status of technophobia and analyze its influencing factors in Chinese older patients with ischemic stroke.

Methods A cross-sectional study was conducted among 204 older patients with ischemic stroke in a tertiary hospital in Tianjin from March 2023 to March 2024. The Technophobia Scale, Electronic Health Literacy Scale (eHEALS), General Self-Efficacy Scale (GSES), and Social Support Rate Scale (SSRS) were used to assess patients' technophobia, electronic health literacy, self-efficacy, and social support, respectively. A self-designed general information questionnaire was used to collect demographic information and disease-related information. Multiple linear regression analysis was performed to analyze the influencing factors.

Results The total score of technophobia in older patients with ischemic stroke was (44.81 ± 10.02). Multiple linear regression analysis showed that education level, monthly income, number of smart devices, electronic health literacy, self-efficacy, and social support were the main influencing factors of technophobia in older patients with ischemic stroke ($P < 0.05$), which could explain 35.3% of the total variation.

Conclusion The level of technophobia of older patients with ischemic stroke deserves attention, suggesting that medical staff can reduce the technophobia of older patients with ischemic stroke according to their sense of self-efficacy, electronic health literacy, and social support, so as to ensure patients' smooth medical treatment.

Clinical Trial Number Not applicable.

Keywords Ischemic stroke, Aged, Technophobia, Influencing factors

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Introduction

Stroke is the first cause of death and disability among adults in China, with five characteristics: high morbidity, high mortality, high disability rate, high recurrence rate, and high economic burden [1]. According to the data from the Centers for Disease Control and Prevention (CDC) of the United States [2], about two-thirds of the hospitalized stroke patients are aged 65 or above. Data from China Stroke Report 2019 showed that the average age of stroke inpatients in China is 65.9 (SD 12.6) years old [3]. Due to the aging population, it is expected that the absolute number of stroke events will increase in the coming decades [4].

Patients with ischemic stroke often have a large number of medical needs, but due to the uneven distribution of medical resources, the functions of patients' bodies are gradually deteriorating, and the accessibility of nursing resources needs to be further improved. The Tenth Five-Year Plan for National Economic and Social Development and the Outline of Long-term Goals in 2035 put forward the implementation of a national strategy to actively respond to the aging population, and emphasized the importance of digital technology in the application of health services for the older people, with "digital transformation as a whole to drive comprehensive changes in production methods, lifestyles, and governance methods [5]. Digitalization and aging are two major trends in current social development, and it is an inevitable requirement for active aging to solve the aging problem by using smart devices and digital information [6]. With the development of digital technology, it can help patients complete appointment registration, health assessment and obtain inspection reports conveniently, and push scientific knowledge of diagnosis and rehabilitation to patients to help them communicate with medical staff online. Digital technology is an important pillar of social development, and individual travel, shopping and medical care are inseparable from digital technology. For the older people, this is undoubtedly a severe challenge, and the change of information technology has undoubtedly promoted the digital divide [5, 6].

With the aging of physical function and memory loss of older patients with ischemic stroke, as well as the influence of cultural level and economic demand, the older patients with ischemic stroke have shown obvious disconnection from digital life in the face of intelligent society [7, 8]. The research shows that the large-scale and high-intensity digital health technology makes the older patients who seek health services feel at a loss and fall into an anxious and nervous state [9, 10]. Due to the limited understanding and acceptance of the older people, the popularization of digital technology also makes them feel at a loss, resulting in a series of negative emotions, and the older patients with ischemic stroke will have certain obstacles in obtaining information due to technical anxiety. Technophobia refers to the irrational anxiety and fear of digital technologies such as mobile communication equipment, artificial intelligence and robots, resulting in emotional reactions such as fear and tension, which will lead individuals to avoid using technology [11].

Tang et al. [12] found social support and electronic health literacy could influence technophobia in older patients with chronic diseases. Peng et al. found a correlation between technophobia and self-efficacy among the older people [13]. In addition, previous studies have identified factors such as education level [12, 14], age [11, 12, 15], monthly income [12, 16], and marital status [12] as being associated with the technophobia. At present, the correlation of electronic health literacy, self-efficacy, and social support with technophobia has not been confirmed in Chinese older patients with ischemic stroke.

Therefore, in this study, a cross-sectional survey was conducted to evaluate the current status of technophobia in older patients with ischemic stroke, and analysed the influencing factors of technophobia, in order to provide reference for improving the technophobia of older patients with ischemic stroke and promoting the development of intelligent old-age care and active aging.

Methods

Study design and participants

Participants were were recruited by convenience sampling from March 2023 to March 2024 in the neurology department of a tertiary hospital in Tianjin. The inclusion criteria for patients were as follows: age ≥ 60 years old; with a clinical diagnosis of ischemic stroke; patients were conscious; with stable vital signs. The exclusion criteria were as follows: patients had language disorders caused by focal lesions in the dominant hemisphere; those who suffered from a major stress event; the mobile phone is not a smart phone.

According to Kendall's [17] sample size calculation method, the sample size is 5 ~ 10 times of the number of variables. In this study, there are 10 items of general data, 3 dimensions of Technophobia Scale, 1 dimension of General Self-Efficacy Scale, 1 dimension of Electronic Health Literacy Scale, and 3 dimensions of Social Support Rate Scale. Considering 20% of the invalid questionnaires, the sample size is calculated as follows: $N = 18 \times 5 \times (1 + 20\%) \sim 18 \times 10 \times (1 + 20\%)$, that is, 108 ~ 216 cases. This study finally collected 204 valid questionnaires.

Data collection

Data collection instruments were a a self-designed general information questionnaire was designed on the basis of consulting literature and group discussion, Technophobia Scale, the Electronic Health Literacy Scale, General Self-Efficacy Scale, and Social Support Rate Scale. All paper questionnaires were distributed by uniformly trained investigators. Before the survey, the survey team was first trained to explain the purpose and significance of this study as well as the precautions when collecting data. First of all, the researcher explained the purpose and significance of the study to the subjects with a unified instruction, and issued a questionnaire with their informed consent, which was filled out by the subjects themselves. For patients without writing or reading ability, the investigators read the items one by one and filled in the records objectively on the spot according to their answers. After the questionnaire was completed, it will be withdrawn and checked on the spot. If there were any omissions or mistakes, the investigator would ask the subjects to fill in and revise them again, and then take them back after confirming them correctly again. For some missing data, interpolation methods such as mean interpolation and multiple interpolation are used to fill in the data. After the invalid questionnaires were eliminated, an Excel database was established and entered by two people, and 10% of the data were randomly selected for review. A total of 210 questionnaires were distributed in this study, of which 204 were valid, and the effective rate was 97.14%.

Table 1 Demographic and Disease-Related characteristics of the study participants and differences of technophobia ($n = 204$)

Variables	*n*(%)	Mean	SD	*F/t*	*P*
Gender				0.517	0.606
female	76 (37.3)	45.09	10.24		
male	128 (62.7)	44.34	9.70		
Age (years)				2.172	0.117
60 ~ 70	96 (47.1)	44.26	10.98		
71 ~ 79	81 (39.7)	46.38	8.22		
≥ 80	27 (13.2)	42.07	10.95		
Education level				7.503	< 0.001
primary school and below	42 (20.6)	48.48	7.57		
junior school	68 (33.3)	46.21	7.14		
high school or technical secondary school	69 (33.8)	43.83	10.46		
college or above	25 (12.3)	37.60	14.62		
Marital status				1.853	0.068
married	152 (74.5)	45.64	9.39		
other	52 (25.5)	42.38	1.43		
Monthly income (RMB)				8.223	< 0.001
< 3000	61 (29.9)	47.77	7.09		
3000 ~ 6000	103 (50.5)	45.01	8.20		
> 6000	40 (19.6)	39.80	15.18		
Place of residence				-0.287	0.775
urban	170 (83.3)	44.72	10.02		
rural	34 (16.7)	45.26	10.18		
Self - care ability				0.611	0.544
mild dependence	65 (31.9)	43.85	12.30		
moderate dependence	111 (54.4)	45.02	8.60		
heavy dependence	28 (13.7)	46.25	9.59		
Course of disease (month)				2.918	0.056
< 3	166 (81.4)	45.39	9.01		
3 ~ 12	32 (15.7)	43.50	12.51		
> 12	6 (2.9)	36.00	17.71		
Number of intelligent devices (unit)				3.708	0.026
0	9 (4.4)	51.56	5.55		
1	170 (83.3)	44.98	9.60		
≥ 2	25 (12.3)	41.28	12.68		

Measurements

The general information questionnaire

Based on the research purpose, a self-designed general information questionnaire was designed on the basis of consulting literature and group discussion. It includes the following two parts: demographic information includes gender, age, education level, marital status, monthly income, residence, and self-care ability; disease-related information includes course of disease and number of intelligent devices. Detailed categorizations of each variable are presented in Table 1. The specific content of the questionnaire can be found in Supplementary Material 1.

Technophobia scale

Technophobia Scale was used to assess technophobia. Technophobia Scale was compiled by Khasawneh [18], the Chinese version of Technophobia Scale was translated and revised by Sun et al. [19], which includes 13 items in 3 dimensions, including techno-anxiety, techno-paranoia, and privacy concerns. Each item was rated on a 5-point Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree). Respondents who scored higher in these three aspects had higher techno-anxiety, techno-paranoia, and privacy concerns. The Cronbach's α coefficient of the total scale is 0.911, and the Cronbach's α coefficients of the three factors are 0.885, 0.836, and 0.759, respectively [18–20]. The specific content of the Technophobia Scale can be found in Supplementary Material 2.

The electronic health literacy scale (eHEALS)

eHEALS was compiled by Norman et al. [21] and translated into Chinese by Guo et al. [22] in 2013 with a total of 8 items and Cronbach's α of 0.913. Each item was rated on a 5-point Likert scale (1 = completely disagree, 5 = completely agree). The total score of the scale ranged from 8 to 40, with higher scores indicating the higher level of electronic health literacy. Studies showed that the Cronbach' s α of the scale was 0.880 ~ 0.958, demonstrating high internal consistency in patients with stroke [7, 8, 21–23]. The specific content of eHEALS can be found in Supplementary Material 3.

General self-efficacy scale (GSES)

GSES is used to evaluate the self-efficacy of patients with chronic diseases [24]. There are 10 items in the scale. Each scored on a scale from 1 to 4. The total score ranges from 10 to 40, with higher scores indicating higher level of self-efficacy, which is suitable for evaluating the self-efficacy level of all kinds of people. GSES has been widely used among older patients with stroke [25–28]. Studies showed that the Cronbach's α of the scale was 0.898 ~ 0.962, indicating high internal consistency. The specific content of GSES can be found in Supplementary Material 4.

Social support rate scale (SSRS)

The scale was compiled by Chinese scholar Xiao [29]on the basis of referring to foreign literature and combining with Chinese national conditions. The scale includes 10 items and the following three dimensions: objective support (3 items), subjective support (4 items) and utilization of support (3 items). The higher the score of the scale, the better the social support. A score of 0 ~ 33 indicates low social support, 33 ~ 45 indicates medium social support, and > 45 indicates high social support. The SSRS has been widely used in 20 disciplines and majors in China. It is also extremely popularly applied among stroke patients, the Cronbach's α was 0.891 ~ 0.920, demonstrating high internal consistency [30–32]. It can effectively assess the social support level of patients, providing a reliable basis for clinical interventions and research. SSRS has the advantages of clear structure, simple operation and strong cultural adaptability, and is suitable for different types of stroke patients [31, 32]. The specific content of SSRS can be found in Supplementary Material 5.

Statistical analysis

SPSS Statistics version 22.0 (SPSS Inc., Chicago, IL, USA) was used for data analysis. The continuous variables conforming to normal distribution are represented by means and standard deviations (SD), categorical variables are represented by n (%). The χ^2 test and independent sample T test were used for univariate analysis. Pearson correlation analysis was used to analyze the correlation between technophobia and electronic health literacy, self-efficacy, and social support in older patients with ischemic stroke, and multiple linear regression analysis was used to analyze the influencing factors of technophobia in older patients with ischemic stroke. Prior to conducting the multiple linear regression analysis, the multicollinearity among the independent variables was carefully examined; the variance inflation factor (*VIF*), tolerance, and Durbin-Watson statistic were calculated to confirm the absence of multicollinearity and to detect the independence of the residuals. The independent variable assignment methods were as following: education level (primary school and below = 1; junior school = 2; high school or technical secondary school = 3; college or above = 4); monthly income (yuan) (< 3000 = 1; 3000 ~ 6000 = 2; >6000 = 3); number of intelligent devices (unit) (0 = 1; 1 = 2; ≥2 = 3); electronic health literacy, self-efficacy, and social support were substituted with the original values. All the tests were bilateral tests, and the difference was statistically significant at $P < 0.05$.

Table 2 Levels of study variables among participants ($n = 204$)

Variables	Mean	SD
Technophobia	44.81	10.02
techno-anxiety	16.96	4.26
techno-paranoia	17.06	3.67
privacy concerns	10.79	2.31
Electronic health literacy	15.82	5.40
Self-efficacy	21.25	4.50
Social support	33.82	3.89
subjective support	18.17	2.76
objective support	9.82	1.02
support utilization rate	5.82	0.64

Results

Characteristics of participants

The majority of participants were male, accounting for 62.7%, mainly 60 ~ 70 years old (47.1%). The research results showed that different educational levels, monthly income, and the number of smart devices were the influencing factors of technophobia of older patients with ischemic stroke, and the differences were statistically significant (all $P < 0.05$). Other demographic and disease-related characteristics are shown in Table 1.

Levels of study variables among participants

Mean ± SD of participants were presented in Table 2.

Correlations of study variables

Table 3 shows Pearson correlations among the study variables. Pearson correlation analysis showed that the correlation coefficients between the total score of technophobia and electronic health literacy, self-efficacy, and social support in older patients with ischemic stroke were − 0.360, −0.328, and − 0.329, respectively, with statistical significance (all $P < 0.01$).

Multiple linear regression analysis of factors related to technophobia in older patients with ischemic stroke

The score of technophobia was defined as the dependent variable, and the independent variables that were statistically significant variables in the univariate

Table 3 Correlation analysis of self-efficacy, electronic health literacy and social support with technophobia in older patients with ischemic stroke

	Technophobia	electronic health literacy	self-efficacy	social support
Technophobia	1			
electronic health literacy	−0.360**	1		
self-efficacy	−0.328**	0.177**	1	
social support	−0.329**	0.184**	0.384**	1

**$p < 0.001$

analysis were entered into the multiple linear regression analysis. The entry level of variables into the equation was $\alpha = 0.05$, and the exclusion level was $\alpha = 0.10$. The tolerance of the regression analysis was 0.805 ~ 0.975 (> 0.01), and the *VIF* was 1.025 ~ 1.242 (≤ 10), confirming the absence of multicollinearity. In addition, the Durbin-Watson value, which was calculated to detect the independence of residuals, was 2.068 (close to 2), indicating that there was no autocorrelation between the error terms. The results of multiple linear regression analysis showed that education level (β=−0.241, 95% CI: −3.800, −1.330; $P < 0.001$), monthly income (β=−0.211, 95% CI: −4.648, −1.401; $P < 0.001$), number of smart devices (β=−0.186, 95% CI: −7.495, −1.785; $P = 0.002$), electronic health literacy (β=−0.290, 95% CI: −0.754, −0.326; $P < 0.001$), self-efficacy (β=−0.138, 95% CI: −0.585, −0.031; $P = 0.030$), and social support (β=−0.178, 95% CI: −0.776, −0.146; $P = 0.004$) are the main influencing factors of technophobia in older patients with ischemic stroke, together explaining 35.3% of the total variance ($R^2 = 0.372$, adjusted $R^2 = 0.353$, $F = 19.369$, $P < 0.001$). The results of multiple linear regression analysis are shown in Table 4.

Discussion

The technophobia of older patients with ischemic stroke is at the upper-middle level

The results of this study showed that the technophobia of older patients with ischemic stroke was at the upper-middle level. The following reasons are considered: the learning and understanding ability of older patients with ischemic stroke declines, and they cannot master new technologies and equipment. In recent years, with the rapid development of information technology for disease prevention and control, medical care procedures are constantly digitized, and a series of information such as appointment registration, health assessment, report inquiry, and rehabilitation information required by patients for medical treatment needs to be submitted or fed back online, which has brought certain medical pressure to patients [33]. Patients will be at a loss but helpless, prone to fear, tension, anxiety and other emotional reactions, leading them to avoid using technology. Coupled with the conservative psychology of older patients and the privacy problems brought about by the development of information technology, patients are obviously worried about information security. With the impact of digital technology, the pressure brought by life and the troubles caused by diseases, the older patients with ischemic stroke are facing multiple tests. When the patients' own abilities are not enough to solve these troubles, all kinds of negative emotions will continue to breed. In this study, the score of technophobia of the older patients with ischemic stroke is (44.81 ± 10.02), which is at the upper-middle level, higher than the result measured by Khasawneh [18]. The reasons may be as follows: the development of smart medical care in the United States is relatively early, and patients are exposed to changes brought about by emerging technologies earlier. Through learning and adaptation, the tension and anxiety in using smart medical service technology are gradually reduced. In China, in March 2019, National Health Commission issued the Notice on Printing and Distributing the Grading Evaluation Standard System of Hospital Smart Services (Trial), which enabled

Table 4 Multiple linear regression analysis of technophobia in older patients with ischemic stroke ($n = 204$)

Variables	B	SE	β	t	*P*	Tolerance	VIF	95% CI Lower	Upper	Durbin-Watson
Constant	96.981	5.935	-	16.342				85.277	108.684	2.025
Education level	−2.565	0.626	−0.241	−4.095		0.925	1.081	−3.800	−1.330	
Monthly income (RMB)	−3.024	0.823	−0.211	−3.674		0.975	1.025	−4.648	−1.401	
Number of intelligent devices (unit)	−4.640	1.448	−0.186	−3.205	0.002	0.952	1.050	−7.495	−1.785	
Electronic health literacy	−0.540	0.108	−0.290	−4.980		0.944	1.059	−0.754	−0.326	
Self-efficacy	−0.308	0.141	−0.138	−2.190	0.030	0.805	1.242	−0.585	−0.031	
Social support	−0.461	0.160	−0.178	−2.884	0.004	0.837	1.195	−0.776	−0.146	

R^2 0.372, adjusted R^2 0.353, *F* 19.369, $P < 0.001$

smart medical care to develop rapidly in China. The development of smart medical care in China is relatively late, and patients' contact time is relatively short, so they have not been able to adapt to and master the technological innovation of smart medical services, so they feel nervous. Nurses and community health service personnel should assess patients' technophobia in time, and analyze the causes of patients' anxiety, tension, and worry, so as to help older patients with chronic diseases better integrate into the smart medical service systems.

Influencing factors of technophobia in older patients with ischemic stroke

Individual factors of patients

This study showed that technophobia in older patients with ischemic stroke was negatively correlated with education level (β=−0.241, $P<0.001$), indicating that the higher level of the education level, the lower the level of technophobia. This may be because older patients with ischemic stroke with higher education levels have higher acceptance of digital health technologies, stronger learning abilities, and higher digital health literacy. During the process of continuous exploration and application, their knowledge and proficiency in using intelligent devices are increasing day by day. Moreover, in this process, their nervousness when using intelligent devices is alleviated. This study showed that technophobia in older patients with ischemic stroke was negatively correlated with the monthly income (β=−0.211, $P<0.001$), indicating that the higher the monthly income, the lower the level of technophobia. This may be because owning digital devices is a prerequisite for the older patients with ischemic stroke to carry out digital practices, and the quality of digital information devices will also affect the enthusiasm of the older patients with ischemic stroke to adapt to the digital ecological environment. Sequelae such as limb dysfunction caused by stroke pose a great challenge to the economic stability of patients' families, especially in developing countries. In addition, the costs associated with the treatment and rehabilitation of ischemic stroke have exerted a great deal of economic pressure on patients [34]. Studies [35, 36] have shown that the income situation of the older population plays a crucial role in the acceptance and adoption of digital health. the older population with lower monthly income is more accustomed to traditional medical methods, often tend to be skeptical of new technologies, and are more likely to experience anxiety and resistance. The older patients with ischemic stroke who have low personal monthly income bear two major economic pressures, namely, daily living expenses and disease treatment. They have few choices in equipment, and some older people use old mobile phones that their children have eliminated. In addition, they are worried that they will have a lot of expenses due to improper operation of products such as smart phones, so they are more prone to technophobia. This suggests that clinical nurses and community workers should not only further strengthen the prevention and control of older patients with ischemic stroke, but also pay more attention to older patients with low education and low income, actively carry out 'digital feedback' from hospitals to communities, strengthen digital education, and help them better integrate into intelligent medical care. This study showed that technophobia in older patients with ischemic stroke was negatively correlated with the number of intelligent devices (β=−0.186, $P=0.002$), indicating that the less intelligent devices, the higher the level of technophobia. This may be because some patients do not have a smartphone, or the quality of their intelligent devices is poor. As a result, they are not proficient in operating intelligent devices, which affects their access to health-related knowledge. On the contrary, older patients with ischemic stroke who own more intelligent devices can have more time and opportunities to get familiar with these devices, which helps them use intelligent devices skillfully and reduces their degree of technophobia to a certain extent. A survey of 1,006 older adults in the eastern region of China showed that the scores of technophobia of the advanced usage group (using the device for ≥3 h per day on average) were significantly lower than those of the beginner usage group (using the device for ≤1 h per day on average), which is directly related to the frequency of device exposure [37]. However, a national survey of urban older people in China has found that moderate use of smartphones (2 ~ 4 h per day on average) can significantly reduce the level of anxiety, but excessive use (≥5 h) may increase the risk of depression. Therefore, future research is needed to explore the relationship between the number of intelligent devices owned by older stroke patients and their level of technophobia. In view of the differences in patients' educational level and understanding, medical institutions at all levels should formulate many measures, such as online medical guidance cards, intelligent voice broadcasting, cyclic video playback, and offline manual service, to ensure the smooth medical treatment of older patients with ischemic stroke. For low-income older patients with ischemic stroke, strategies could include providing low-cost or free smartphones or tablets through government subsidies, public welfare donations, or collaborating with enterprises to launch "aging-friendly packages" (e.g., simplified devices) to reduce technophobia caused by device shortages or high costs.

Electronic health literacy

In this study, the electronic health literacy of the older patients with ischemic stroke was negatively correlated with the level of technophobia (β=−0.290, $P<0.001$). The higher the total score of electronic health literacy, the lower the level of technophobia, which is consistent with the research results of Arcury et al. [38]. The reason for analyzing this may be related to the fact that technophobia may hinder patients from utilizing digital health resources, which in turn affects their rehabilitation process and health status [39]. Older patients with ischemic stroke with lower electronic health literacy have relatively weak abilities to understand and evaluate digital information technology and lack the ability to actively use new technologies. For older patients with ischemic stroke who have a low level of electronic health literacy, when they passively come into contact with or have to use digital technologies, their sense of unfamiliarity with the technologies and their inability to operate them will significantly intensify their feelings of nervousness and fear. On the other hand, older ischemic stroke patients with higher electronic health literacy are more proficient in obtaining health information through smart devices, conducting remote consultations, and managing diseases independently. They are more likely to recognize the effectiveness of online health information in improving their own health [40]. They are also more willing to obtain health-promoting resources and knowledge from the Internet, strengthen their awareness of health management, and practice health-promoting behaviors, thereby reducing the degree of technophobia [41]. Therefore, improving the electronic health literacy of older patients with ischemic stroke has become a key breakthrough point for alleviating technological anxiety and facilitating their integration into the digital medical system.

Self-efficacy

In this study, the self-efficacy of the older patients with ischemic stroke was negatively correlated with the level of technophobia (β=−0.318, $P=0.030$), indicating the higher the total score of self-efficacy, the lower the level of technophobia. The reason for analyzing this may be related to the fact that higher self-efficacy can help older patients with ischemic stroke effectively improve their disease adaptability and health status, and improve their disease self-management level. The self-efficacy level of patients with ischemic stroke is of paramount importance in their rehabilitation process. A higher level of self-efficacy enables them to effectively deal with psychological problems such as anxiety and depression, which plays a positive role in their rehabilitation [42, 43]. Compared with younger groups, the older people have a lower acceptance of new things. When faced with smart health technologies, they often lack confidence, which makes their self-efficacy relatively low when using related applications [20]. Tao et al. [44] found that the higher the self-efficacy, the stronger the perceived ease of use and the more willing to use Internet medical care. Older people with higher self-efficacy often have stronger adaptability and are able to better master and apply new technologies. This enables them to form more positive health concepts, which also alleviates their fear of technology to a certain extent [20]. Medical staff should cultivate the self-efficacy of older patients with ischemic stroke, build a communication platform between doctors and patients by using the Internet, share the knowledge related to stroke and the experience of disease self-management, and encourage family members to participate in the process of patient disease management, so as to further enhance older patients' confidence in coping with the disease and fully mobilize their subjective initiative to actively learn network information.

Social support

In this study, the social support of the older patients with ischemic stroke was negatively correlated with the level of technophobia (β=−0.178, $P=0.004$) indicating the higher the total score of social support, the lower the level of technophobia. This is consistent with the results of a study by Luo et al. [20] showing that social support is a direct negative predictor of technophobia in older adults. The reason for analyzing this may be related to the fact that stable economic support, good life experience, and effective psychological support are more conducive to improving the level of mindfulness in older patients with ischemic stroke [43, 45]. After realizing the convenience and practicality of digital health technology, the acceptance of digital health technology is also higher, and the level of technophobia of older patients with ischemic stroke becomes lower. It is suggested that the younger generation should accompany the older patients to seek medical treatment as far as possible. Medical institutions at all levels can set up navigation nursing posts for older patients in outpatient clinics to provide navigation services for patients in need, and provide one-to-one guidance services when necessary.

Implications for clinical practice

Through investigation and analysis, this study identified the main factors influencing technophobia among older patients with ischemic stroke. The findings of this study highlight the need to develop effective strategies to alleviate technophobia and, eventually, improve the quality of life in older patients with ischemic stroke. First, for older patients with ischemic stroke who have difficulties understanding technical terms and have weak logical reasoning, it is recommended to adopt visual and scenario-based teaching methods to avoid relying on text. The uses of technology can be explained in combination with daily life scenarios, and paper operation manuals with rich pictures and text descriptions should be provided. These manuals should be accompanied by 'device operation flowcharts' and posted in prominent positions at home to reduce the memory burden on patients. Additionally, the operation interface can be simplified, including the development of 'one-click' function modules to reduce patients' cognitive load, and replacing complex text input with voice interaction. Second, government departments, medical insurance departments, etc. should provide subsidies for equipment purchases and waive data traffic fees for low-income patients. This can also help patients who have fewer devices. Finally, it is recommended that medical staff cooperate with senior universities and senior activity centers to incorporate the use of smart devices into regular courses and design relevant scenarios in combination with rehabilitation needs. Enterprise resources can also be introduced to hold 'technology open days', allowing older patients with ischemic stroke to personally experience suitable devices. What's more, peer support groups should be established, enabling technologically confident elderly people to guide beginners and making full use of the principles of social learning. In conclusion, addressing technophobia in older patients with ischemic stroke requires a multifaceted approach that integrates psychological support, technological innovation, and system adaptation. By integrating these strategies into clinical workflows, healthcare providers can increase patients' engagement with digital health tools, ultimately improving the rehabilitation outcomes and quality of life of older patients with ischemic stroke.

Limitations and recommendations for future study

However, this study has several limitations. First, participants were recruited from a hospital using a convenient sampling method and may not be sufficiently representative of the population, thus limiting the generalizability of the results. Its generalizability may not be fully applicable to a broader older population or patients in different medical environments. Second, there are individual differences in the causes of technophobia of older patients with ischemic stroke, and further research is needed to explore its promoting factors and obstacles, and to formulate practical individualized intervention measures to alleviate their technophobia. Third, this study is a cross-sectional study design, the research can only reveal the correlations between variables, making it difficult to establish causal relationships. Technophobia is a psychological barrier. Relying solely on statistical data for interpretation may be too simplistic, and it is unable to comprehensively and deeply explore the deeper influencing factors that affect the technological anxiety of older patients with ischemic stroke, such as sociocultural and economic factors. In the future, qualitative research or mixed-methods research will be carried out to deeply explore the essence and connotations of the phenomenon of technophobia among older patients with ischemic stroke, as well as the more profound influencing factors. The government should increase financial investment in the construction of network service system for the older patients, support and encourage them to use information technology from the policy aspect, and help the older patients with ischemic stroke to obtain health information.

Conclusions

In conclusion, the technophobia of older patients with ischemic stroke can not be ignored. Education level, monthly income, number of smart devices, electronic health literacy, self-efficacy, and social support were the main factors affecting technophobia of older patients with ischemic stroke. According to the influencing factors of technophobia of older patients with ischemic stroke, it is suggested that the service of "internet plus Smart Pension" should be actively carried out at the social level, such as policy support, resource integration, technological research and development, and build a complete service system to effectively meet the older people's pension needs. Furthermore, electronic information equipment suitable for older patients with ischemic stroke should be developed to meet their special needs when using new technologies.

Abbreviations

eHEALS	Electronic Health Literacy Scale
GSES	General Self-Efficacy Scale
SSRS	Social Support Rate Scale
SD	Standard Deviations
VIF	Variance Inflation Factor

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12877-025-06304-x.

Supplementary Material 1.

Acknowledgements

The authors are grateful to all the patients who participated in the survey.

Authors' contributions

Y. S. was a major contributor to the writing of the manuscript. Y. S. and R. Z. composed the proposal, participated in data collection, and revised the manuscript. D. H. and H. M.participated in the study design, data collection, manuscript revision, and supervised the whole study. C. W. and Y. W. contributed to the literature search and wrote part of the draft. All authors have read and agreed to the published version of the manuscript.

Funding

This study was sponsored by Tianjin Health Research Project (No. TJWJ2022QN036), Research Project of Tianjin Union Medical Center (No. 2024YJHL029).

Data availability
All the data generated or analyzed during this study are included in this published article. The datasets used and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate
This study was approved by the Medical Ethics Committee of Tianjin Union Medical Center in accordance with the principles of the Declaration of Helsinki. The ID for ethics approval is (2024) Expedited Review No. (B97). An informed consent was obtained from all participants before data collection. In the whole research process, we strictly abide by the principles outlined in the Helsinki Declaration to ensure the anonymity and confidentiality of participants' information and data.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

Received: 27 November 2024 / Accepted: 18 July 2025
Published online: 19 August 2025

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